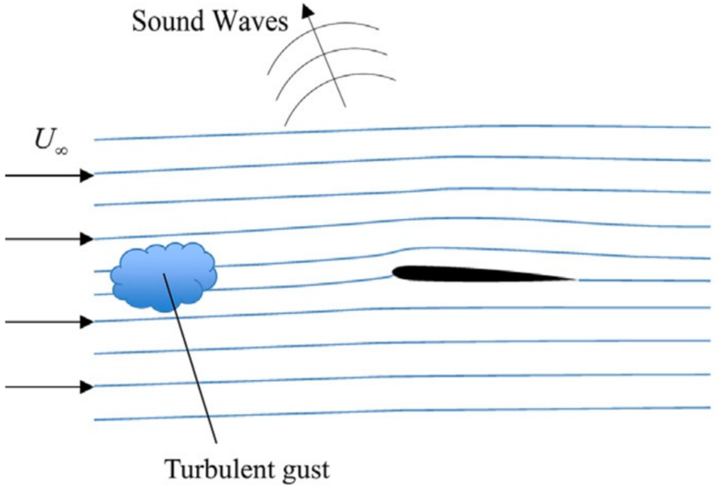
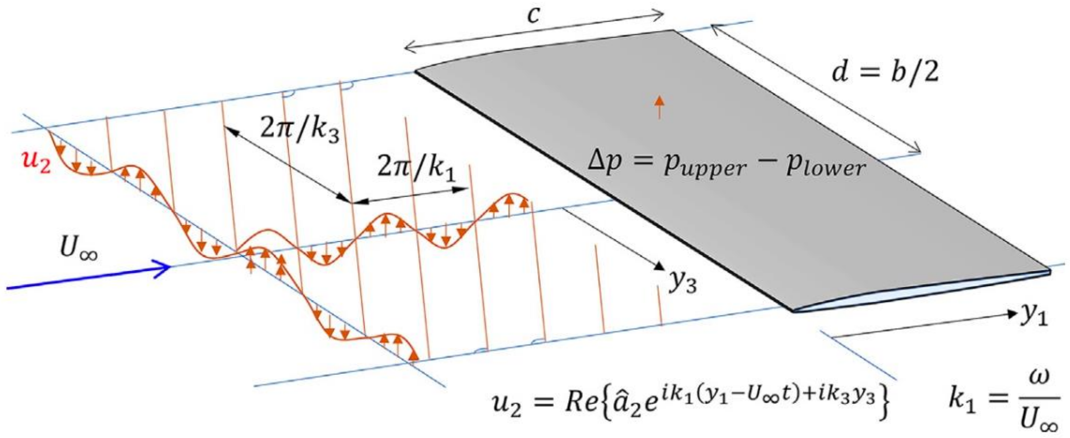
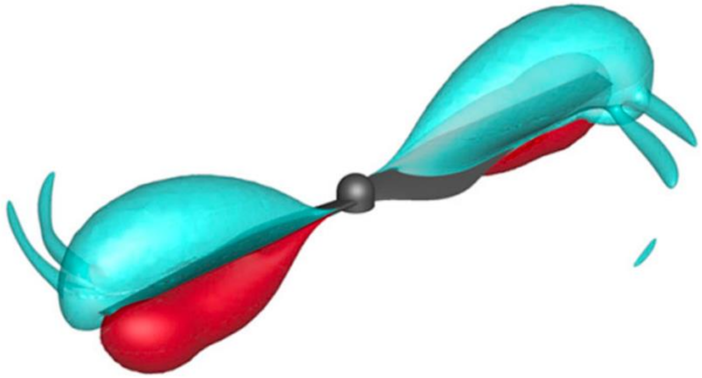
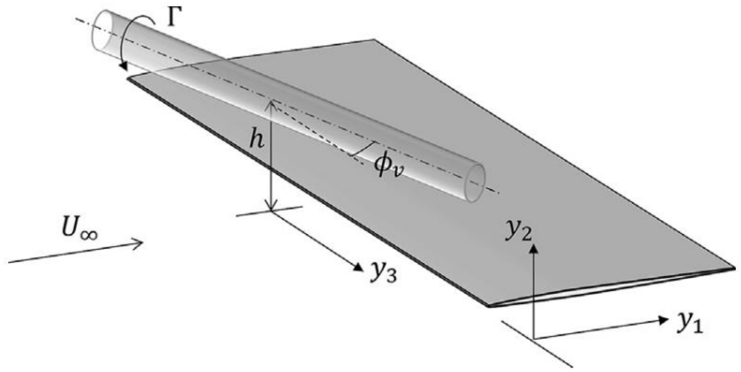




Chapter 4

Unsteady Loading

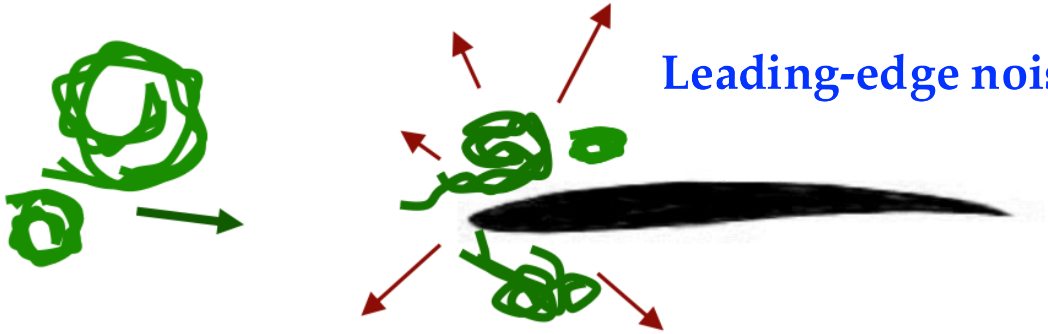




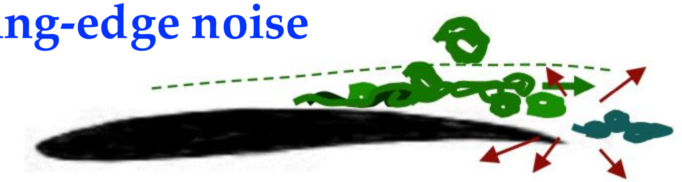
Unsteady loading noise

- Unsteady pressure fluctuations induced on an aerodynamic surface (unsteady loading) by a turbulent (random) field can be generated due to:

Leading-edge noise

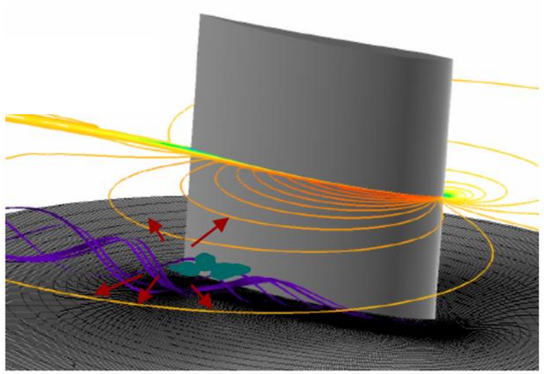


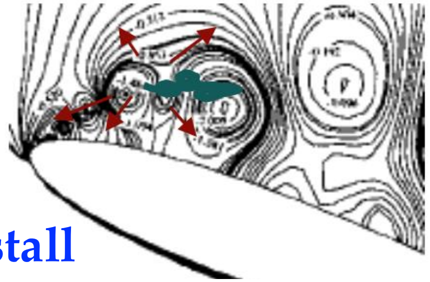
Trailing-edge noise



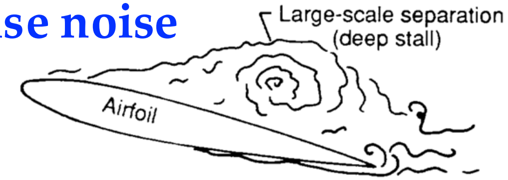
$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}(\nabla \times \mathbf{u})} \Rightarrow \textbf{noise}$$

Junction noise

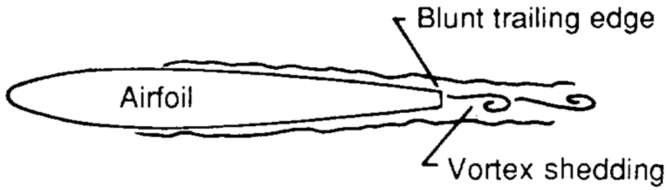




Separation stall noise noise



Vortex shedding noise



Unsteady loading noise

- Recall the **far-field approximation** of a **dipole source** for a **thin airfoil** with negligible viscous effects:

$$\boxed{(\tilde{p}(\mathbf{x},\omega)c_\infty^2)_{\text{dipole}} \approx \frac{-i\pi\omega x_2 \Delta\tilde{\tilde{p}}(k_1^{(o)},k_3^{(o)},\omega)e^{ik|\mathbf{x}|}}{c_\infty|\mathbf{x}|^2}}$$

$$\mathbf{k}^{(o)} = (k_1^{(o)},k_2^{(o)},k_3^{(o)}) = k\frac{\mathbf{x}}{|\mathbf{x}|} = \frac{\omega}{c_\infty}\frac{\mathbf{x}}{|\mathbf{x}|}$$

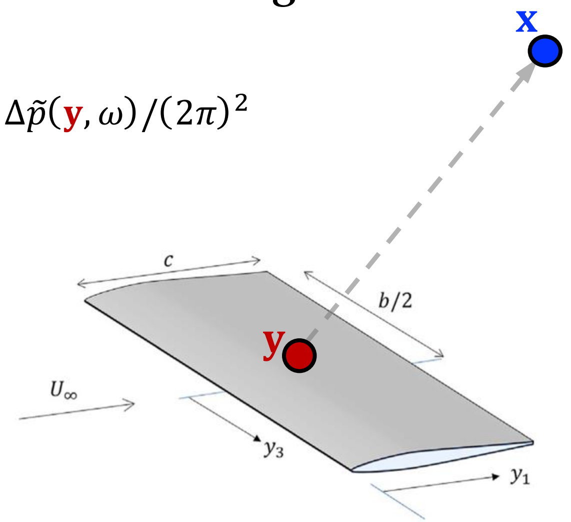
Acoustic *wavevector* in the observer direction

2D *wavenumber frequency transform* (chordwise-spanwise) of the pressure jump over the airfoil planform

$$\Delta\tilde{\tilde{p}}(k_1,k_3,\omega) = \frac{1}{(2\pi)^2}\int_{-R_\infty}^{R_\infty}\int_{-R_\infty}^{R_\infty}\Delta\tilde{p}(\mathbf{y},\omega)e^{-ik_1y_1-ik_3y_3}dy_1dy_3$$

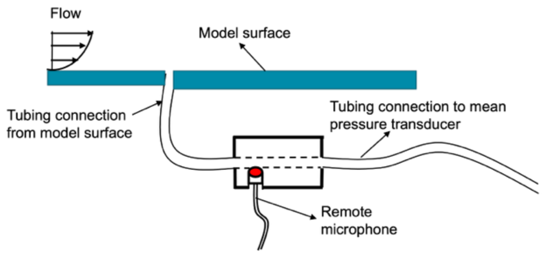
- For a given ω and $\mathbf{x}/|\mathbf{x}|$, the **far-field acoustic pressure** is proportional $\Delta\tilde{\tilde{p}}$ evaluated at a *single* wavenumber pair $(k_1^{(o)},\ k_3^{(o)})$ with pressure disturbnances convecting at c_∞
  - At **low frequencies**, $k_i^{(o)} \to 0$, so $\Delta\tilde{\tilde{p}}(k_1^{(o)},k_3^{(o)},\omega)$ reduces to the net unsteady loading on the airfoil, $\Delta\tilde{p}(\mathbf{y},\omega)/(2\pi)^2$
  - At **very high frequencies**, $\Delta\tilde{\tilde{p}}$ will be dominated by **edge effects** as the *integrand's oscillatory nature* tends to *suppress* the contributions from the surface where pressure varies smoothly

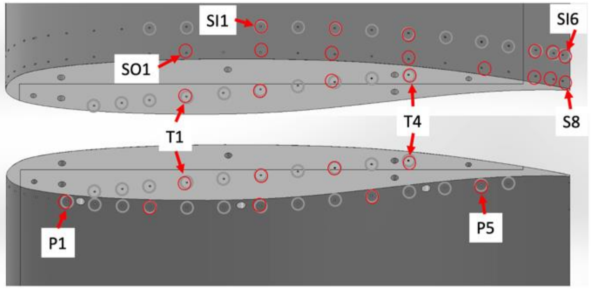
For airfoils – we can assume the behavior along the chord (y_1) is imposed by **2𝔻 theories of unsteady aerodynamics** (e.g., Sears and Amiet extension), often assuming ***statistical homogeneity*** along the span (y_3)

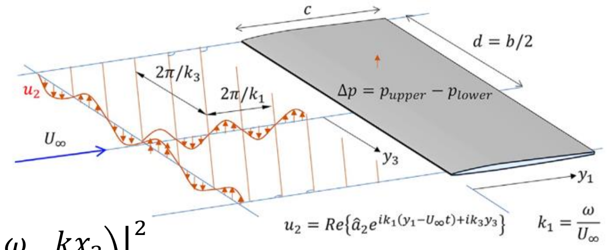


Unsteady loading noise

- Several methods are possible to assess the fluctuating surface pressure spectra:
  1. **Direct measurement** – using appropriate instrumentation
  2. **Empirical modeling** – based on experiment or working hypotheses
  3. **Physical modeling using analytical formulations** – feasible only in simple configurations (e.g., Sears theory and Amiet's extension)
  4. **Numerical simulation** – complex and often expensive (e.g., LES, DNS)

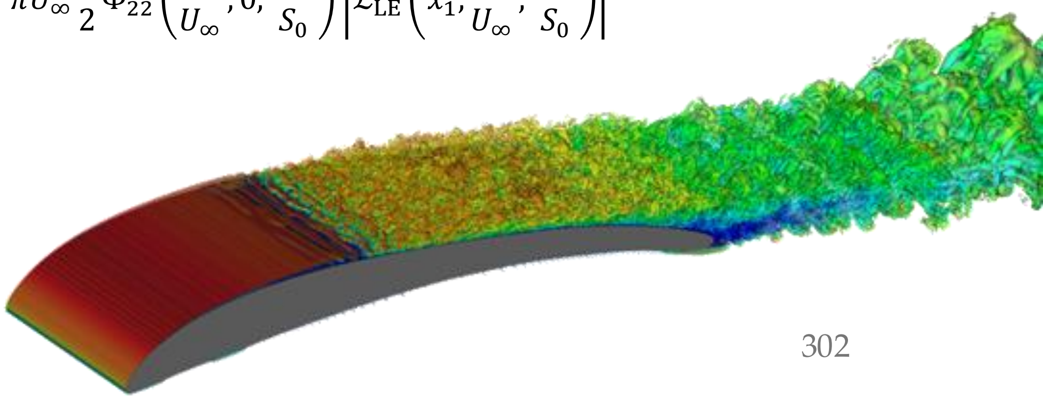






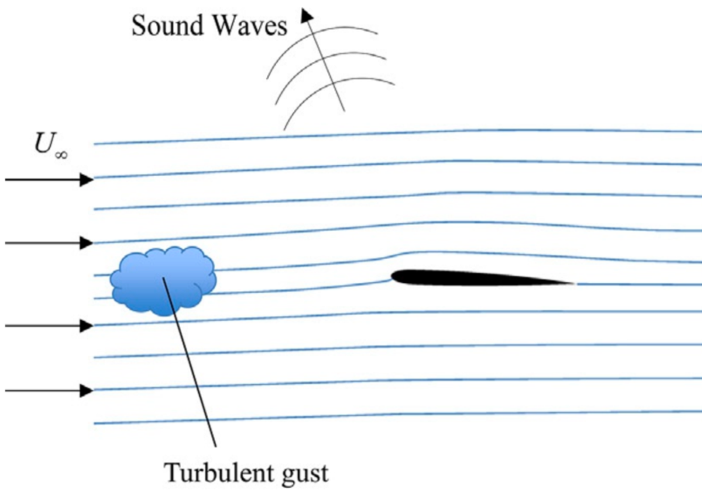
$$S_{pp}^{\mathrm{TE}}(\mathbf{x},\omega) = \left(\frac{\omega c x_2}{2\pi c_\infty S_0^2}\right)^2 \frac{b}{2} \Phi_{pp}(\omega)\, l_z\left(\frac{k x_3}{S_0},\omega\right) \left|\mathcal{L}_{\mathrm{TE}}\left(x_1,\frac{\omega}{U_\infty},\frac{k x_3}{S_0}\right)\right|^2$$

$$S_{pp}^{\mathrm{LE}}(\mathbf{x},\omega) = \left(\frac{\rho_\infty \omega c x_2}{2 c_\infty S_0^2}\right)^2 \pi U_\infty \frac{b}{2} \Phi_{22}\left(\frac{\omega}{U_\infty},0,\frac{k x_3}{S_0}\right) \left|\mathcal{L}_{\mathrm{LE}}\left(x_1,\frac{\omega}{U_\infty},\frac{k x_3}{S_0}\right)\right|^2$$

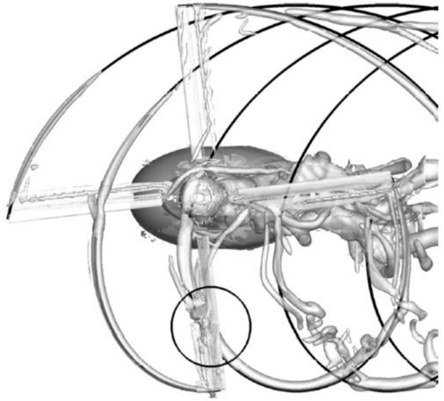


Amiet's approach

- After laying the foundation of aeroacoustics, we now describe approaches to analyze the unsteady fluid dynamics needed to estimate **unsteady surface loading**
  - The dominant source at low Mach flows, and particularly for **rotors**
- We begin by analyzing the unsteady fluid dynamics of a thin airfoil in a uniform stream, **ignoring viscous effects** and assuming **small disturbances** (linearized Euler eq.)
  - Basis to analyze the response of an airfoil to free stream disturbances in incompressible & compressible flow
  - Sound prediction is done by integrating the solution into Curle's eq. or FW-H eq.
- We will see that the **incompressible Sears function solution** would apply only when the **airfoil is a compact noise source**
- A more sophisticated compressible analysis will be carried out for all non-compact applications



Amiet's approach



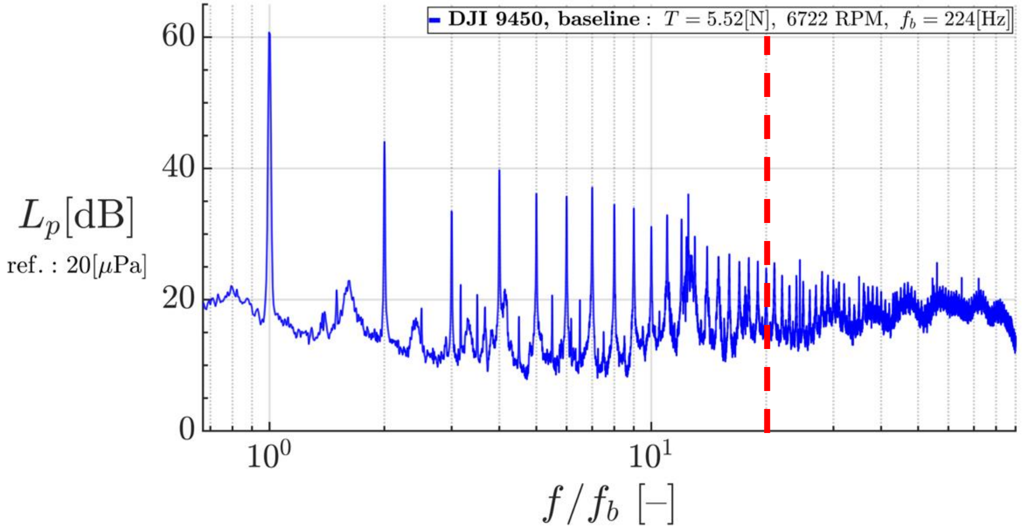
- ❑ Sound generated by propellers/rotors largely depended on the **unsteady blade loading**, generated as the blades move through a region of *nonuniform flow*
- ❑ Consider the case of an *incompressible flow*, which implies that the blade chord c is **acoustically compact**:

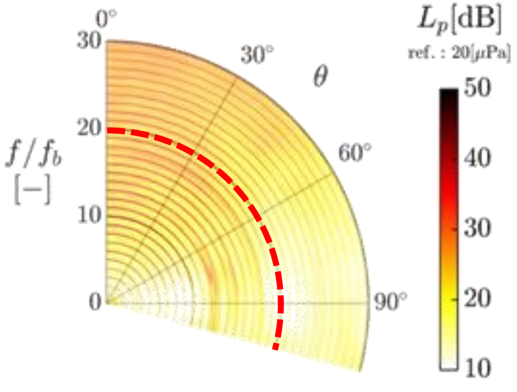
$$\frac{c}{\lambda} = c\frac{f_b}{c_\infty} = c\frac{mB}{c_\infty}\frac{\omega}{2\pi} \implies \frac{c}{\lambda} = \left(\frac{c}{R}\right)\left(\frac{\omega R}{c_\infty}\right)\left(\frac{mB}{2\pi}\right) \implies \boxed{\frac{c}{\lambda} = \mathrm{AR}^{-1}\, M_{\mathrm{tip}}\left(\frac{mB}{2\pi}\right)}$$

f_b – blade passing frequency, $mB\omega/2\pi$ [Hz]
ω – angular speed of the blades [rad/s]
m – harmonic of blade passing frequency
B – no. of blades

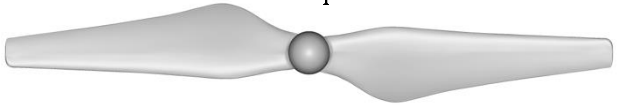
- ➢ A DJI 9450 rotor is considered acoustically compact up to its $m = 20$ harmonic! ($c/\lambda < 0.25$)

$$\left(\frac{c}{\lambda}\right)_{\mathrm{DJI}} = 0.012m$$





$B = 2$
$R = 120$ mm
$S_{\mathrm{ref}} = 2{,}141$ mm^2
$\mathrm{AR} = R^2/S_{\mathrm{ref}} = 6.73$
$M_{\mathrm{tip}} = 0.25$



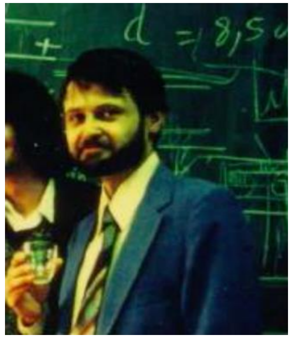
Amiet's approach

- The incompressible flow mechanism that causes the unsteady loads on a rotor blade when it passes through a region of nonuniform flow is a *straightforward extension of the potential flow theory*
- Recall **potential flow** result for lift per unit span L' of a 2𝔻 thin symmetric airfoil:
  - *Assumes* incompressible, irrotational 2𝔻 flow, where thickness/camber are ignored (flat plate)
  - α – angle of attack; Γ – circulation, $= \oint_C \mathbf{v} \cdot d\mathbf{l}$

$$L' = -\rho U_\infty \Gamma$$

$$\Gamma = -\pi U_\infty c \sin\alpha$$

- **Roy K. Amiet** was among the first to extend the above and provide practical methods for the prediction of **leading- and trailing-edge noise** for the case of a **thin airfoil in a uniform stream**[1-3]



Roy K. Amiet
July 19, 1942 – Oct. 12, 2024

Journal of Sound and Vibration (1975) **41**(4), 407–420

ACOUSTIC RADIATION FROM AN AIRFOIL IN A TURBULENT STREAM

R. K. Amiet

United Aircraft Research Laboratories, East Hartford, Connecticut 06108, *U.S.A.*

(*Received* 8 *October* 1974, *and in revised form* 13 *February* 1975)

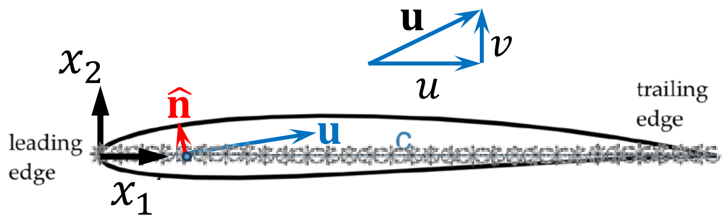
A theoretical expression for the far-field acoustic power spectral density produced by an airfoil in a subsonic turbulent stream is given in terms of quantities characteristic of the turbulence. For an observer directly above the airfoil the relevant quantities are the spanwise correlation length of the turbulence as a function of frequency and the PSD of the vertical velocity fluctuations, both quantities being measured in an airfoil fixed co-ordinate system. In the derivation it is assumed that the spanwise correlation length is much smaller than the airfoil span.

A more solid theoretical foundation is developed for a lift expression given by Liepmann for an airfoil in turbulence. Also, the analysis shows why it is not necessary, within certain limitations, to integrate over all gust wavenumbers in order to calculate the sound at a given observer location. Rather, for small turbulence length scales the observer hears only the sound produced by a particular wavenumber component of the turbulence, and thus the analysis can be simplified considerably.

A comparison of the theory with experimental acoustic and lift measurements available in the literature shows good agreement.

[1] R. K. Amiet (1975) "Acoustic radiation from an airfoil in a turbulent stream", J. Sound Vib. 41, pp. 407–420.
[2] R. K. Amiet (1976) "High frequency thin-airfoil theory for subsonic flow", AIAA J. 14(8), pp. 1076–1082.
[3] R. K. Amiet (1976) "Noise due to turbulent flow past a trailing edge", J. Sound Vib. 47(3), 387–393.

Amiet's approach



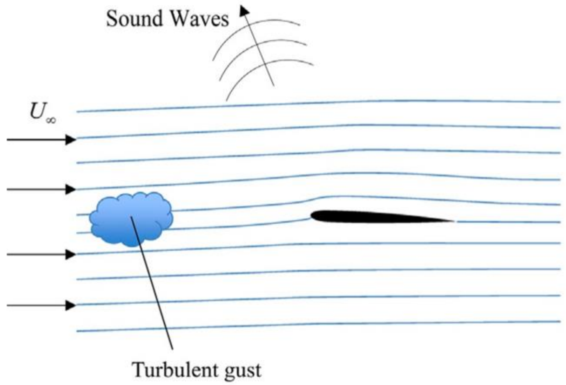
- Recall ***thin airfoil assumptions*** (based on small disturbances):
  1. Small α; such that $\sin\alpha \approx \alpha$ and $\cos\alpha \approx 1$
  2. Small velocity changes along the x-direction, $U \gg u(\mathbf{x},t)$
  3. Neglect airfoil thickness, camber, and camber slope; BC imposed @ $x_2 = \pm 0$ rather than at the actual surface
- Consider a flat plate in a uniform flow $\mathbf{U} = (U, V, 0)$, such that $\alpha = \tan^{-1}(\mathrm{V/U})$
- Unsteady flow region causes speed to change: $\mathbf{U} + \mathbf{u}(\mathbf{x},t) = (U + u(\mathbf{x},t), V + v(\mathbf{x},t), 0)$
- **Angle of attack at the blade LE** is: $\alpha = \tan^{-1}\left(\dfrac{V + v(\mathbf{x}_{\mathrm{LE}},t)}{U + u(\mathbf{x}_{\mathrm{LE}},t)}\right)$

$\alpha_{t=0} = 0 \mid V = 0$

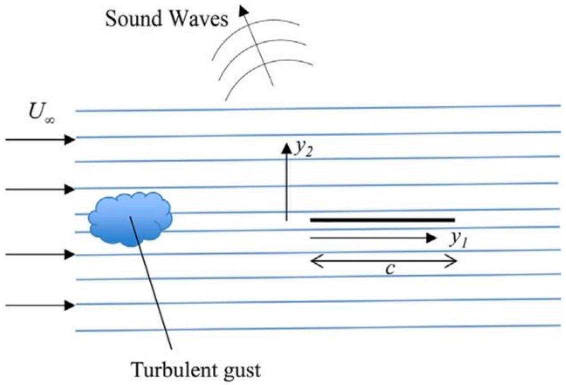
$U \gg u(\mathbf{x},t)$

$$\boxed{\alpha(t) \approx \tan^{-1}\left(\frac{v(\mathbf{x}_{\mathrm{LE}},t)}{U}\right)}$$

$$\boxed{L'(t) = \pi c \rho U_\infty v(\mathbf{x}_{\mathrm{LE}},t)}$$



Simplify



Amiet's approach

- **Problem definition: determine the unsteady loading on the surface as the gust is convected past the airfoil (flat plate)**
- Recall the *momentum equation* with negligible viscous effects ($\sigma_{ij} \approx 0$) and ***small*** density perturbations compared to the mean ($\rho' \ll \rho_o$)

$$\frac{\partial(\rho v_i)}{\partial t} + \frac{\partial(\rho v_i v_j + p_{ij})}{\partial x_j} \approx \rho_o \frac{\partial v_i}{\partial t} + \frac{\partial(p_o + p')}{\partial x_i} + \rho_o \frac{\partial(v_i v_j)}{\partial x_j} \approx 0$$

$$\rho = \rho_o + \rho'$$
$$p = p_o + p'$$
$$\mathbf{v} = \mathbf{U} + \mathbf{u}(\mathbf{x}, t)$$

- Velocity in above equation is the mean flow plus velocity perturbation ($v_i = U_i + u_i$):

$$\rho_o \left(\frac{\partial u_i}{\partial t} + U_j \frac{\partial u_i}{\partial x_j} + u_j \frac{\partial U_i}{\partial x_j} + \cancel{U_i \frac{\partial u_j}{\partial x_j}} + \cancel{u_i \frac{\partial U_j}{\partial x_j}} \right) + \cancel{\frac{\partial(p_o + p')}{\partial x_i}} + \rho_o \frac{\partial\left(\cancel{U_i U_j} + \cancel{u_i u_j}\right)}{\partial x_j} \approx 0$$

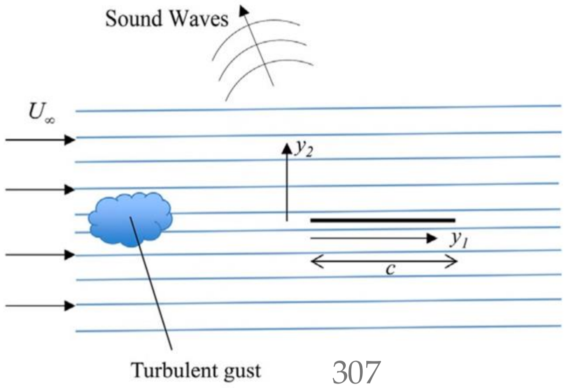
$U_i \gg u_i$

$\partial v_i / \partial x_i = 0$

Averaging the momentum eq. yields:
$\partial p_o / \partial x_i = -\rho_o \partial(U_i U_j) / \partial x_j$

Linearized Euler equation for unsteady perturbations about the mean flow

$$\boxed{\rho_o \left(\frac{\partial u_i}{\partial t} + U_j \frac{\partial u_i}{\partial x_j} + u_j \frac{\partial U_i}{\partial x_j} \right) + \frac{\partial p'}{\partial x_i} \approx 0}$$



Amiet's approach

$$\rho_o\left(\frac{\partial u_i}{\partial t}+U_j\frac{\partial u_i}{\partial x_j}+u_j\frac{\partial U_i}{\partial x_j}\right)+\frac{\partial p'}{\partial x_i}\approx 0$$

- **For a flat plate in uniform flow:**

$$\mathbf{U}=(U_\infty,0,0): \quad \boxed{\rho_o\frac{D_\infty u_i}{Dt}+\frac{\partial p'}{\partial x_i}\approx 0} \qquad \text{where:} \quad \frac{D_\infty u_i}{Dt}=\frac{\partial u_i}{\partial t}+U_\infty\frac{\partial u_i}{\partial x_1}$$

- To separate the gust from the flow induced by the airfoil to match the impermeable BC on the surface, we will use a **Helmholtz decomposition** for the unsteady flow component **u**:

$$\boxed{\mathbf{u}=\nabla\phi+\mathbf{u}^{(r)}}$$

irrotational field (compressible)

rotational field (incompressible)

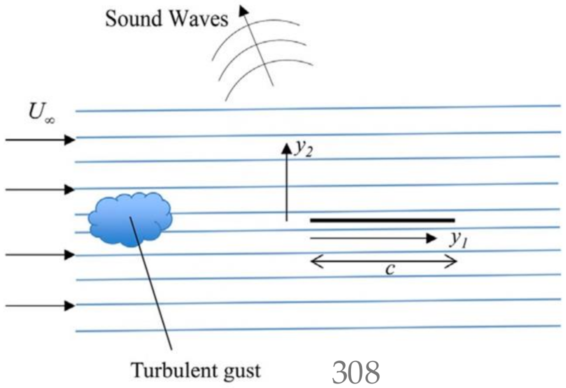
$$\mathbf{u}^{(r)}=\nabla\times\boldsymbol{\psi}$$

$$\nabla\times\nabla\phi=0 \quad ; \quad \nabla\cdot(\nabla\times\boldsymbol{\psi})=0$$

$$\begin{cases}\nabla\cdot\mathbf{v}=\nabla^2\phi\\ \nabla\times\mathbf{v}=-\nabla^2\boldsymbol{\psi}\end{cases}$$

- Substituting the above decomposition into the **momentum eq.** yields:

$$\nabla\left(\rho_o\frac{D_\infty\phi}{Dt}+p'\right)+\rho_o\frac{D_\infty\mathbf{u}^{(r)}}{Dt}\approx 0$$



$$\frac{D_\infty}{Dt} = \frac{\partial}{\partial t} + U_\infty \frac{\partial}{\partial x_1}$$
$$\nabla \times \nabla \phi = 0$$
$$\nabla \cdot (\nabla \times \boldsymbol{\psi}) = 0$$

Amiet's approach

- Taking the curl of the momentum equation leads to:

$$\nabla\left(\rho_o \frac{D_\infty \phi}{Dt} + p'\right) + \rho_o \frac{D_\infty \mathbf{u}^{(r)}}{Dt} \approx 0 \quad \overset{\nabla \times ()}{\Longrightarrow} \quad \frac{D_\infty \omega_i}{Dt} = 0$$

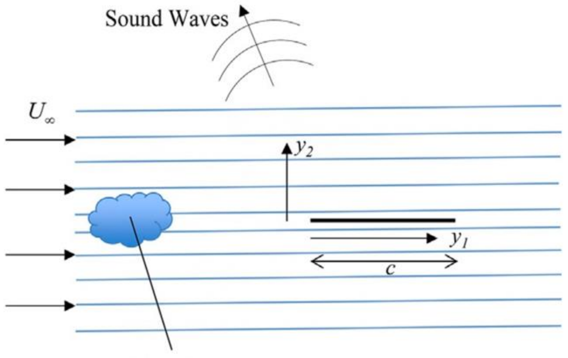
ω_i - vorticity of the perturbation velocity

  - Vorticity must be convected by the mean flow, so it is a function of $\omega_i(x_1 - U_\infty t, x_2, x_3)$

- The rotational part of the gust $u_i^{(r)}$ comes from the vorticity, so it must evolve similarly $\Longrightarrow \frac{D_\infty \mathbf{u}^{(r)}}{Dt} = 0$
  - Therefore, both terms of the momentum eq. are identically zero !

$$\nabla\left(\rho_o \frac{D_\infty \phi}{Dt} + p'\right) \approx 0 \quad \Longrightarrow \quad \boxed{p' = -\rho_o \frac{D_\infty \phi}{Dt}} \quad + \text{ arbitrary const.}$$

  - Thus, **potential flow disturbances define the pressure perturbations in the flow**



$$\frac{D_\infty}{Dt} = \frac{\partial}{\partial t} + U_\infty \frac{\partial}{\partial x_1}$$

$$\rho = \rho_o + \rho'$$
$$p = p_o + p'$$
$$\mathbf{v} = \mathbf{U} + \mathbf{u}(\mathbf{x}, t)$$

Amiet's approach

- We shall also satisfy **continuity equation**, which can be written as follows:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho v_j)}{\partial x_j} = 0 \quad \Longrightarrow \quad \frac{\partial \rho}{\partial t} + v_j \frac{\partial \rho}{\partial x_j} + \rho \nabla \cdot \mathbf{v} \ = 0 \quad \Longrightarrow \quad \frac{\partial \rho'}{\partial t} + \cancel{u_j \frac{\partial \rho'}{\partial x_j}} + U_j \frac{\partial \rho'}{\partial x_j} + (\rho_o + \cancel{\rho'})\nabla \cdot \mathbf{u} = 0$$

  - Keep only first-order terms

$$\Longrightarrow \quad \frac{D_\infty \rho'}{Dt} + \rho_o \nabla \cdot \mathbf{u} = 0$$

Using isentropic relation $p' = c_\infty^2 \rho'$

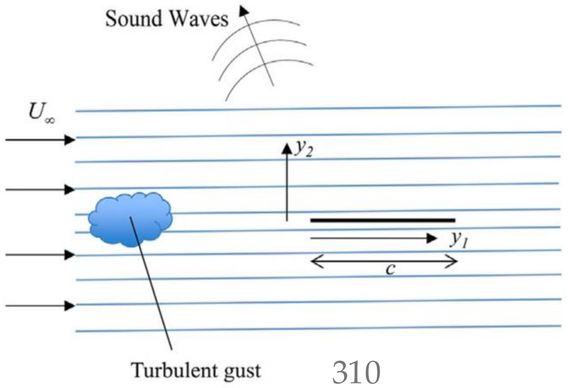
$$\boxed{\frac{1}{c_\infty^2}\frac{D_\infty p'}{Dt} + \rho_o \nabla \cdot \mathbf{u} = 0}$$

- We substitute the expression for p' we obtained from the momentum eq.:

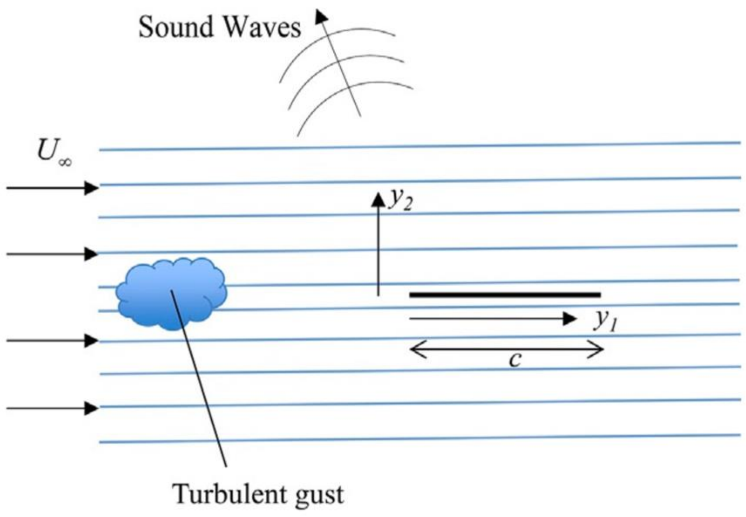
$$\frac{1}{c_\infty^2}\frac{D_\infty}{Dt}\left(-\rho_o \frac{D_\infty \phi}{Dt}\right) + \rho_o \nabla \cdot \mathbf{u} = 0 \quad \Longrightarrow \quad -\frac{\rho_o}{c_\infty^2}\frac{D_\infty}{Dt}\left(\frac{D_\infty \phi}{Dt}\right) + \rho_o \nabla^2 \phi = 0$$

$$\boxed{\frac{1}{c_\infty^2}\frac{D_\infty^2 \phi}{Dt^2} - \nabla^2 \phi = 0}$$

Convected Wave Equation



Amiet's approach



$$\boxed{\frac{1}{c_\infty^2}\frac{D_\infty^2\phi}{Dt^2} - \nabla^2\phi = 0}$$

- The equation shows that an **upstream incompressible gust is convected by the mean flow (at U_∞) without distortion**
- The **airfoil surface introduces** an **additional velocity disturbance** that is represented by $\mathbf{u} = \nabla\phi$, **satisfying the convected acoustic wave eq.**
- When $c_\infty \to \infty$ (*incompressible flow*), we obtain Laplace's eq.: $\nabla^2\phi = 0$
- **For the case of an airfoil**, the mean flow may not be uniform, so the **upstream gust** is no longer simply convected by the mean flow and is ***stretched* by mean flow acceleration and velocity gradients**

Flat plate case (*undistorted mean flow*) **provides a 1st order estimate for the unsteady blade loading**
to compute rotor noise, as pioneered by Amiet…

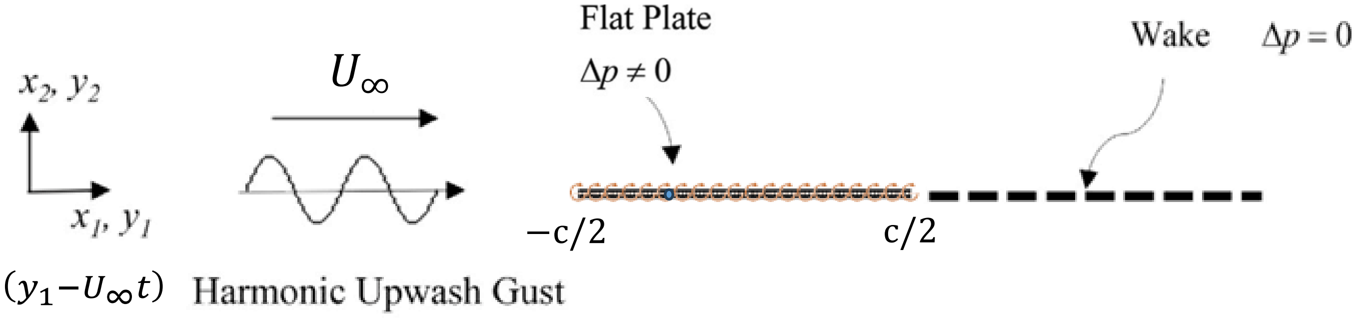
von Kármán and Sears problem

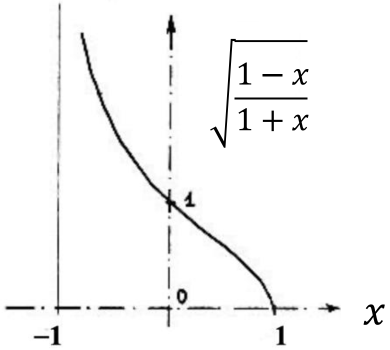
Incompressible flow

- **Unsteady harmonic upwash gust encountering a flat plate (thin airfoil) of chord c at $\alpha = 0°$**
  - Thin airfoil theory assumptions apply (*small* perturbations, $U \gg u$, *negligible* camber/thick)
  - Gust causes a change in angle of attack ($\Delta\alpha$) → changes circulation around the plate ($\Delta\Gamma$)
    → results in vorticity shedding to the wake such that Kelvin's theorem is satisfied at every instant ($D\Gamma/Dt = 0$)
    - Kutta condition is satisfied at every instant by considering vorticity in the wake to convect with U_∞
  - Later, Amiet expressed a more 'correct' BC by requiring that $\Delta p = 0$ across the (*thin 'free-force'*) wake
  - Since vorticity is convected downstream at speed close to that of the flow, there is a **delay (phase lag) between the lift response and the upstream velocity fluctuations** → Sears function (lift deficiency)

$$\boxed{L'(t) = \mathbb{R}\mathrm{e}\left[\pi c \rho_\infty U_\infty \hat{a}_2 S(\sigma) e^{-ik_1 U_\infty t}\right]}$$

$$\sigma = \frac{k_1 c}{2}$$ *Non-dimensional frequency*



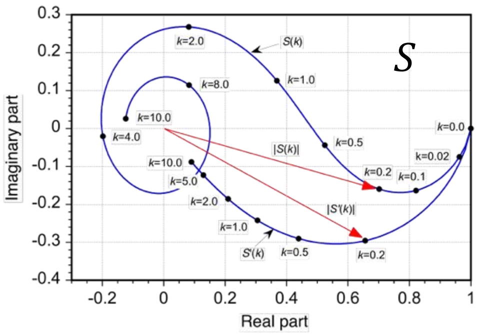


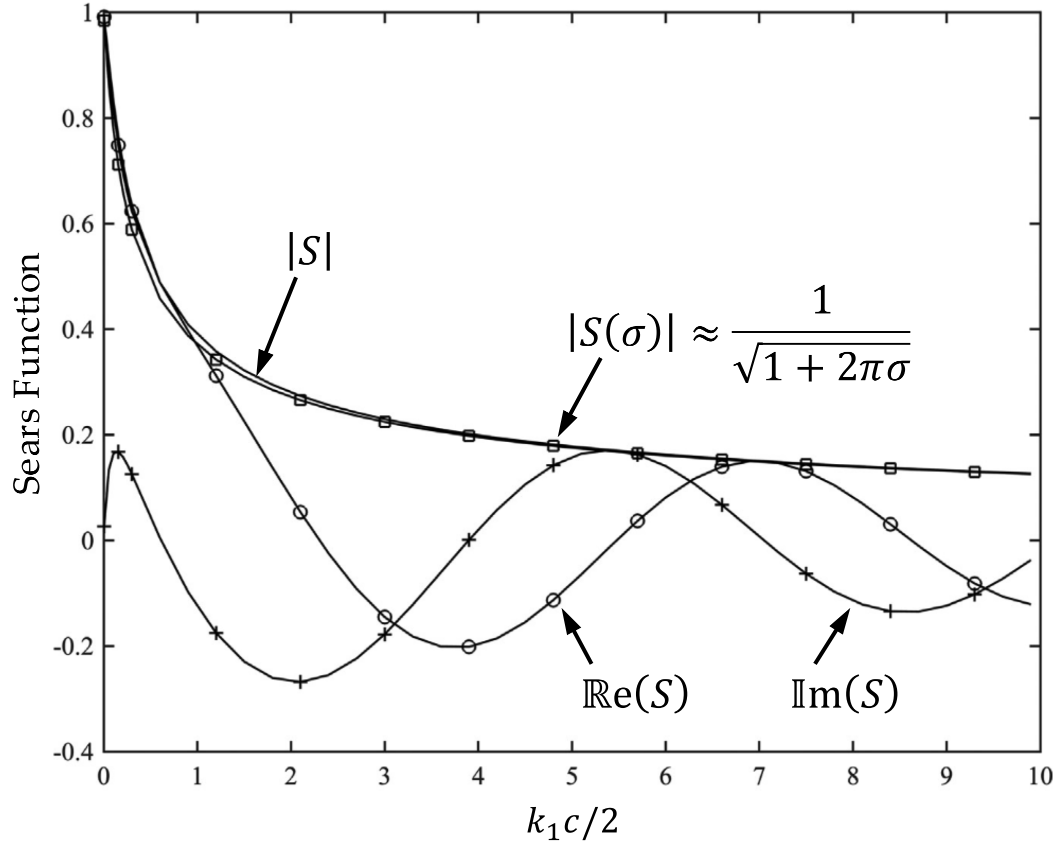
$$k_1 = \frac{\omega}{U_\infty} \qquad u_2 = \hat{a}_2 e^{ik_1(y_1 - U_\infty t)}$$

[1] T. von Kármán and W. R. Sears (1938) "Airfoil theory for non-uniform motion", J. Aeronautical Sciences 5(10), 379–390.

von Kármán and Sears problem
Incompressible flow

- **Sears function**
  - Amiet found it **valid for acoustic frequencies of $\lambda > 4c$** – *low frequency approximation*





$$L'(t) = \mathbb{Re}\left[\pi c \rho_\infty U_\infty \hat{a}_2 S(\sigma) e^{-ik_1 U_\infty t}\right]$$

$$\boxed{S(\sigma) = \frac{2}{\pi\sigma\left(H_0^{(1)}(\sigma) + iH_1^{(1)}(\sigma)\right)}}$$

$$\sigma = \frac{k_1 c}{2}$$

Non-dimensional frequency

$$\lim_{\sigma\to 0} \sigma H_0^{(1)}(\sigma) = 0 \qquad \lim_{\sigma\to 0} \sigma H_1^{(1)}(\sigma) = -2i/\pi$$

MATLAB implementation

$H_\nu^{(1)}(z)$ = $besselh(\nu, 1, z)$

Computes Hankel function of the first kind $H_\nu^{(1)}(z) = J_\nu(z) + iY_\nu(z)$

$J_\nu(z)$ – Bessel function of the first kind
$Y_\nu(z)$ – Bessel function of the second kind

von Kármán and Sears problem
Incompressible flow

- Recall that for a **rigid stationary surface** that is **acoustically compact**, the **far-field dipole** noise can be approximated as:

$$\boxed{\rho'(\mathbf{x},t)c_\infty^2 \approx \frac{x_i}{4\pi|\mathbf{x}|^2 c_\infty}\left[\frac{\partial F_i}{\partial \tau}\right]_{\tau=\tau^*}}$$

- By substituting Sears' solution, we obtain the **acoustic far field pressure** in terms of the upwash gust amplitude for an observer outside the flow:

$$F_2(\tau^*) = -L = -\mathbb{R}\mathrm{e}[\pi c b \rho_\infty U_\infty \hat{a}_2 S(\sigma)]e^{-ik_1 U_\infty \tau^*}$$

$\tau^* = t - |\mathbf{x}|/c_\infty$

b – span

$$\boxed{p'(\mathbf{x},t) \approx \mathbb{R}\mathrm{e}\left(\frac{-i\omega x_2 e^{-i\omega(t-|\mathbf{x}|/c_\infty)}}{4\pi|\mathbf{x}|^2 c_\infty}[\pi c b \rho_\infty U_\infty \hat{a}_2 S(\sigma)]\right)}$$

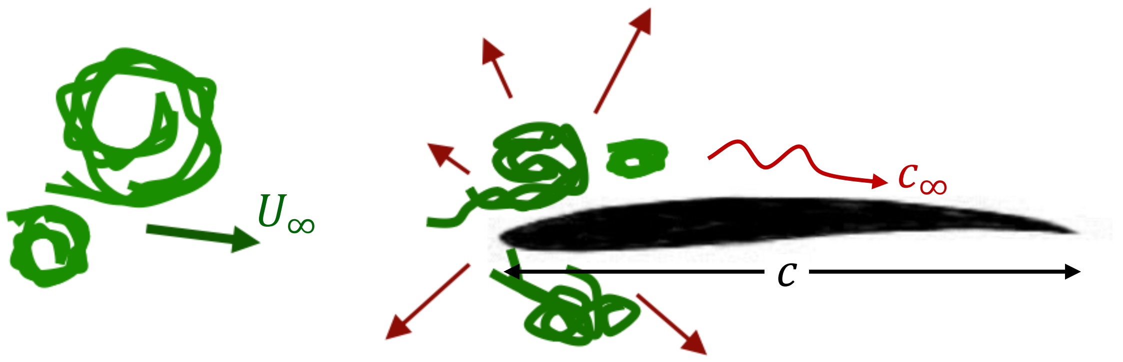
- This is an acoustic dipole that points in the direction of the unsteady lift, with a cosine directionality (no sound in the direction of the flow)

Compressible flow response to a step gust

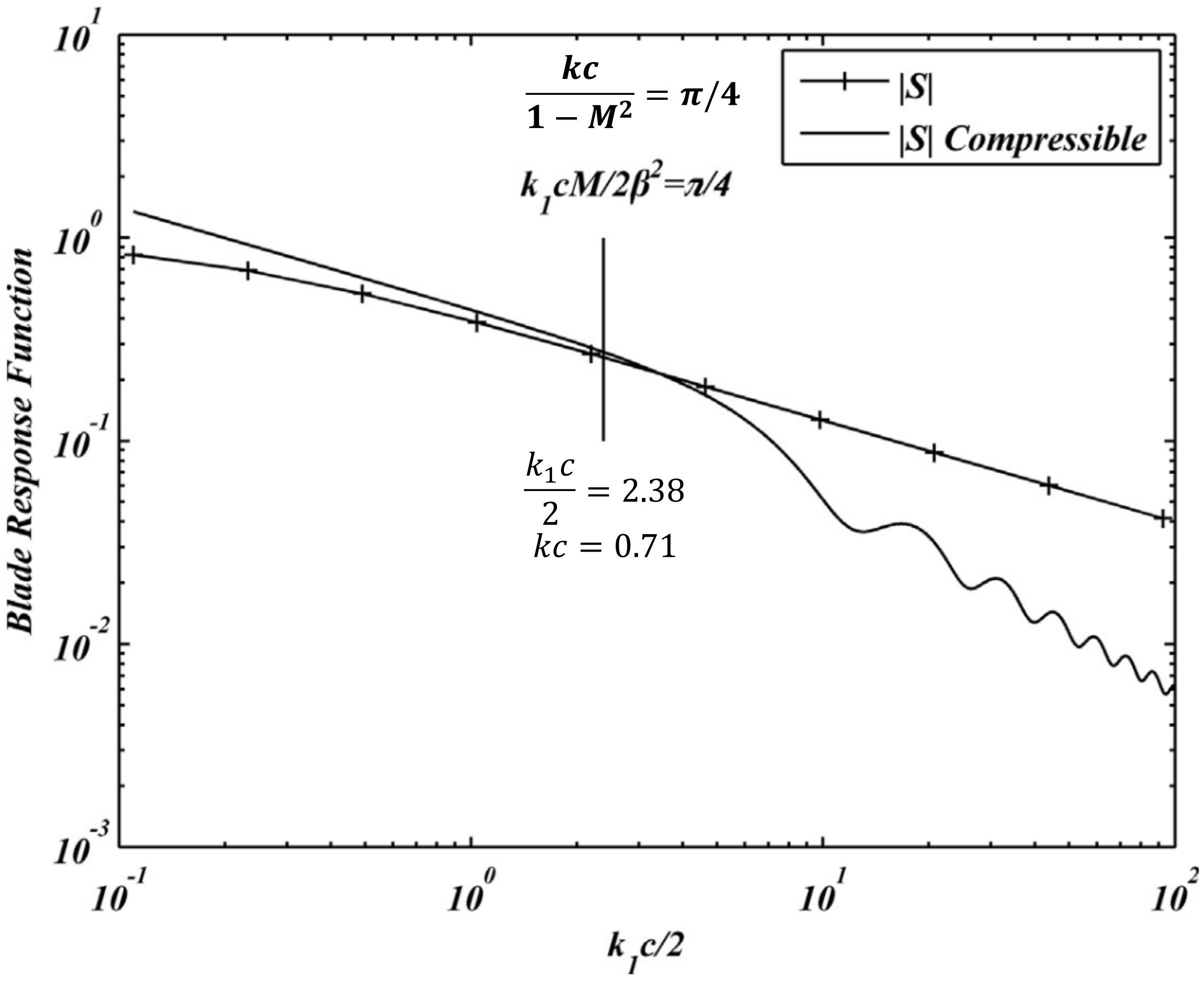
- **Incompressible flow** – whole flow reacts *instantaneously* to any change in BCs, and so when the gust strikes the LE, vorticity is shed into the wake at the same instant to satisfy the Kutta condition at the TE
- **Compressible fluid** – information propagates through the medium at c_∞, and so the **BC at the TE will be unaltered until the acoustic wave generated at the LE by the gust interaction reaches the TE**
  - Time taken for the acoustic wave to travel from the LE to the TE is $T_{\mathrm{LE}\rightarrow\mathrm{TE}} = c/(c_\infty + U_\infty)$
  - To assess the delay importance, we compute the phase shift between LE pressure fluctuations & the TE pressure, which for a frequency of interest ω, is $\omega c/(c_\infty + U_\infty)$, or $kc/(1+M)$
  - For incompressible flow to be valid $\Longrightarrow \dfrac{kc}{1+M} \ll 1$

For example, a frequency of 1000 Hz and a chord of 0.30 m at $M = 0$, implies $kc = 5.5$, and so the *incompressible flow assumption* is far from valid…

Only relevant for low frequencies!



Compressible flow response to a step gust



Compressible blade response function compared to Sears function as a function of the non-dimensional frequency $\sigma = k_1c/2$, for a low Mach number of $M = 0.3$

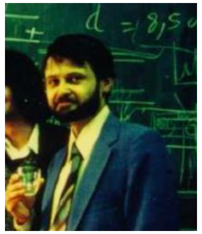
Amiet's leading- and trailing-edge solutions
Overview

- Amiet[1-3] addressed the problems of **leading- and trailing-edge noise** by using the solution to the **Schwartzschild** problem, which was developed for electromagnetic wave scattering in the presence of a semi-infinite half-plane
  - Solution to Schwartzschild problem is derived using the **Wiener-Hopf method** (Sec. 20.2.2, Glegg & Devenport) and **only applies to a stationary surface in the absence of flow** (extension to uniform flow will follow)

Helmholtz equation

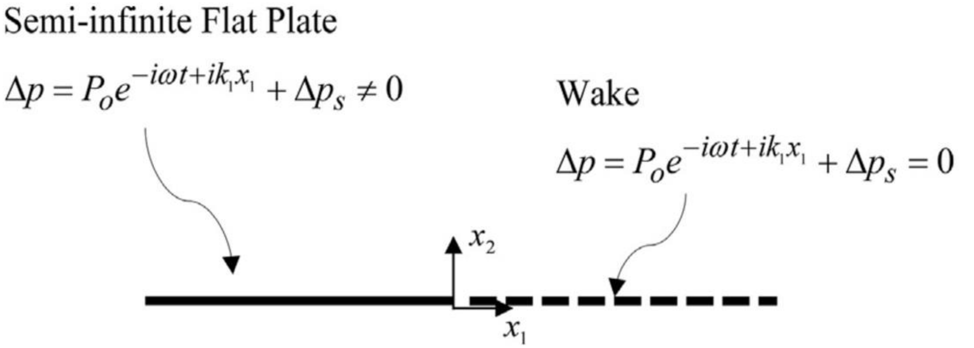
$$\boxed{\frac{\partial^2 \hat{p}}{\partial x_1^2} + \frac{\partial^2 \hat{p}}{\partial x_2^2} + k^2 \hat{p} = 0}$$

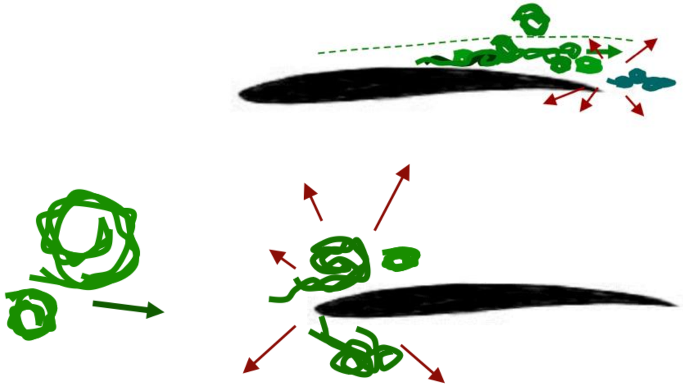
BC:
$$\begin{cases} [\Delta \hat{p}_s(x_1)]_{x_1 \geq 0} = -P_o e^{ik_1 x_1} & \textit{Kutta} \\ \left[\dfrac{\partial \hat{p}(x_1,0)}{\partial x_2}\right]_{x_1<0} = 0 & \textit{Impermeable} \end{cases}$$



Roy K. Amiet
July 19, 1942 – Oct. 12, 2024

Karl Schwartzschild
Oct. 9, 1873 – May 11, 1916





[1] R. K. Amiet (1976) "Noise due to turbulent flow past a trailing edge", J. Sound Vib. 47, pp. 387–393
[2] R. K. Amiet (1978) "Effect of incident surface pressure field on noise due to turbulent flow past a trailing edge", J. Sound Vib. 57, pp. 305–306.
[3] R. K. Amiet (1976) "High frequency thin airfoil theory for subsonic flow", AIAA J. 14, 1076–1082.

Amiet’s leading- and trailing-edge solutions

Schwartzschild problem and TE solution

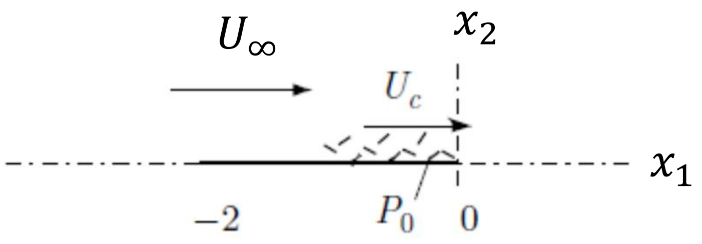
- Schwartzschild solved the problem to give the pressure jump over the plate ($x_1 < 0$):

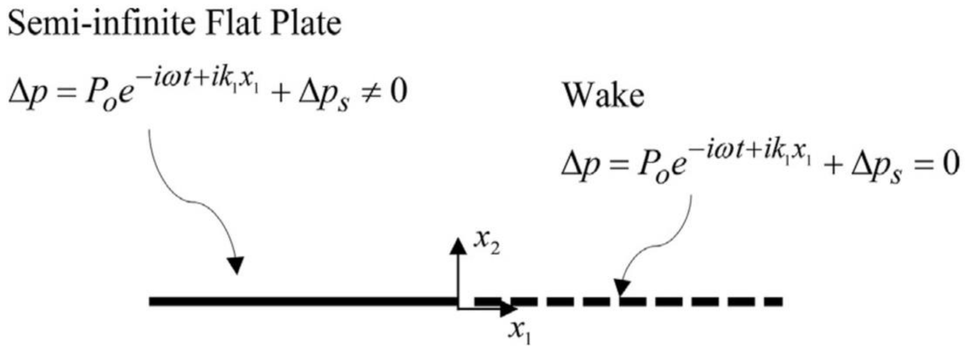
$$\boxed{\Delta p_s(x_1) = -\frac{2P_o}{\pi}\int_0^{\infty}\sqrt{\frac{|x_1|}{\xi}}\,\frac{e^{ik(\xi+|x_1|)+ik_1\xi}}{\xi+|x_1|}\,d\xi} \qquad x_1 < 0$$

  - **The generation of sound is interpreted as a linear wave diffraction mechanism**

- Evaluating the integral above results in:

$$\boxed{\Delta\hat{p}_s(x_1) = P_o e^{ik_1x_1}\left[(1-i)E_2\big((k+k_1)|x_1|\big)-1\right]} \qquad x_1 < 0$$



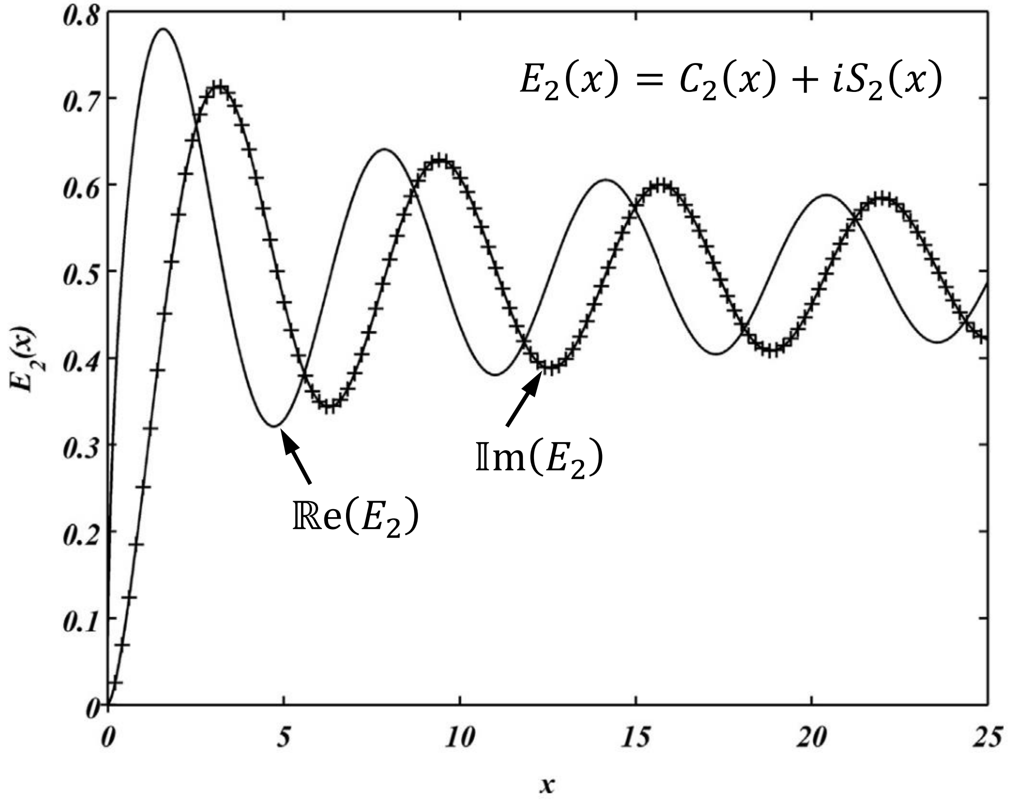


$E_2(x)$ - **modified Fresnel integral**

$$\boxed{E_2(x) = \int_0^x \frac{e^{i\zeta}}{\sqrt{2\pi\zeta}}\,d\zeta}$$

Amiet's leading- and trailing-edge solutions

Schwartzschild problem and TE solution



$E_2(x)$ - **modified Fresnel integral**

$$\boxed{E_2(x) = \int_0^x \frac{e^{i\zeta}}{\sqrt{2\pi\zeta}} d\zeta}$$

MATLAB implementation of $E_2(x)$:

```
fresnelc(sqrt (2*x/pi)) + i*fresnels(sqrt(2*x/pi))
```

- $E_2(x)$ appears often in the theory of edge scattering in several different forms

Amiet's leading- and trailing-edge solutions

Effect of uniform flow on Schwartzschild TE solution

- The Schwartzschild problem must be modified to account for the **mean flow over the surface**
- Also, since surface pressure perturbation can be three-dimensional, we extend our 2𝔻 analysis to 3𝔻
- Thus, the pressure and velocity potential perturbations must satisfy the **convective wave** eq.:

$$\boxed{\frac{1}{c_\infty^2}\frac{D_\infty^2 p'}{Dt^2} - \nabla^2 p' = 0} \qquad \text{where:} \quad p' = -\rho_o \frac{D_\infty \phi}{Dt}$$

  - Time derivative is expressed in the frame of reference of an observer who is moving relative to the medium

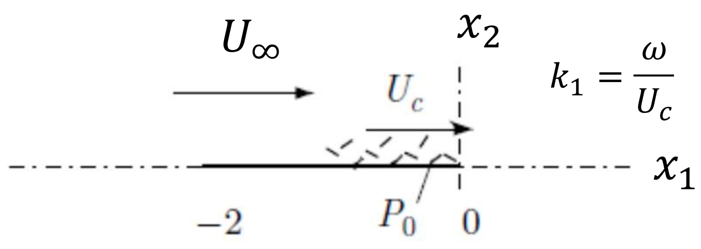
- Surface pressure perturbation $\Delta\hat{p}_s$ is considered harmonic, but is convected downstream in the direction of the flow, and can include a harmonic *spanwise variation*, so that the BC is modified to:

$$[\Delta\hat{p}_s(x_1, x_3)]_{x_1 \geq 0} = -P_o e^{ik_1 x_1 + ik_3 x_3}$$

- Incident disturbance is harmonic in time and the surface geometry is independent of the spanwise direction, thus:

$$[\hat{p}(x_1, x_2, x_3)]_{3\mathbb{D}} = [\hat{p}(x_1, x_2)]_{2\mathbb{D}} e^{ik_3 x_3}$$

$$\Longrightarrow \quad p'(\mathbf{x}, t) = [\hat{p}(x_1, x_2)]_{2\mathbb{D}} e^{-i\omega t + ik_3 x_3}$$



Amiet's leading- and trailing-edge solutions

Effect of uniform flow on Schwartzschild TE solution

- Substituting into the convective wave equation, we obtain:

$$\boxed{\beta^2 \frac{\partial^2 \hat{p}}{\partial x_1^2} + 2ikM \frac{\partial \hat{p}}{\partial x_1} + \frac{\partial^2 \hat{p}}{\partial x_2^2} + (k^2 - k_3^2)\hat{p} = 0}$$

$$k = \frac{\omega}{c_\infty} \qquad \beta^2 = 1 - M^2 \qquad M = \frac{U_\infty}{c_\infty}$$

- Solving the above results in the blade response to an incident pressure disturbance, which can later be used to evaluate trailing-edge noise:

$$\boxed{\Delta\hat{p}_s(x_1, x_3) = P_o g_{\mathrm{TE}}(x_1, k_1, k_3) e^{ik_3 x_3}} \qquad x_1 < 0 \qquad A = 1 + \frac{\kappa + k_o M}{k_1}$$

$$\boxed{g_{\mathrm{TE}}(x_1, k_1, k_3) = e^{ik_1 x_1}[(1-i)E_2(k_1 A|x_1|) - 1]} \qquad k_o = \frac{\omega}{\beta^2 c_\infty} \qquad \kappa^2 = k_o^2 - \frac{k_3^2}{\beta^2}$$

- When $k_3 = 0$ and $U_c = U_\infty$: $A = \dfrac{1}{1-M} > 0$
  - Mean flow effect has a factor of $1/(1-M)$, resulting in a response more rapid close to $x_1 = 0$

NOTE!
κ can be imaginary for large values of k_3!
Yet, for $\Delta\hat{p}_s$ that couple to the acoustic far field κ must always be real!

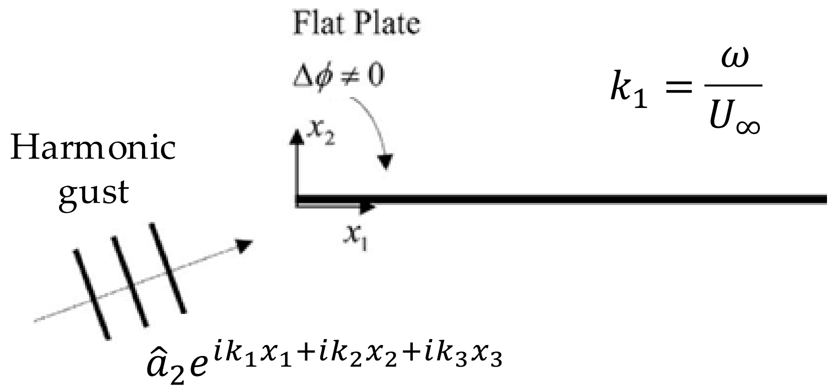
Amiet's leading- and trailing-edge solutions

LE solution

- Effect of a **finite chord** is essential to solve the LE scattering problem due to an unsteady upwash gust
- Thus, we need to include the effect of both the LE and the TE in the solution, as follows:
  1. Solve the problem of an upwash gust incident on a semi-infinite flat plate
  2. Add a TE correction based on the Schwartzschild solution described previously to ensure that there is no pressure jump across the wake downstream of the blade TE
     - Correction introduces a small jump in potential upstream of the LE, so additional terms need to be added to correct this (this is a relatively small correction that can usually be ignored)
- The convective wave eq. is defined with the velocity potential perturbation : $\boxed{\frac{1}{c_\infty^2}\frac{D_\infty^2\phi}{Dt^2} - \nabla^2\phi = 0}$

where: $\hat{\phi}(x_1, x_2, x_3)e^{-i\omega t} = \hat{\phi}(x_1, x_2)e^{-i\omega t + ik_3x_3}$

$$\textbf{BC:} \begin{cases} \left[\Delta\hat{\phi}(x_1, x_3)\right]_{x_1<0} = 0 \\ \left[\dfrac{\partial\hat{\phi}(x_1, 0, x_3)}{\partial x_2} + \hat{a}_2 e^{ik_1x_1 + ik_3x_3}\right]_{x_1>0} = 0 \end{cases}$$



Amiet's leading- and trailing-edge solutions
LE solution

- Solving the LE problem results in the semi-infinite plate response to an unsteady upwash gust:

$$\boxed{\Delta\hat{p}^{(1)}(x_1,x_3) = \pi\rho_\infty\hat{a}_2 U_\infty g^{(1)}(x_1,k_3,\omega,M)e^{ik_3x_3}}$$

$$\beta^2 = 1 - M^2 \qquad M = \frac{U_\infty}{c_\infty}$$

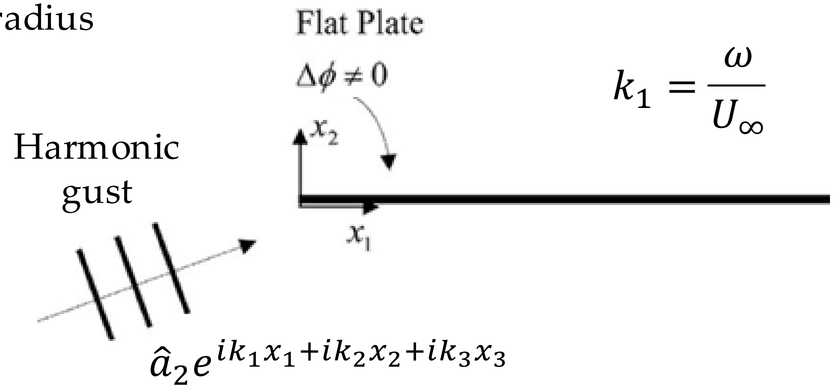
$$\boxed{g^{(1)}(x_1,k_3,\omega,M) = \frac{-2e^{i(\kappa - Mk_o)x_1 + i\pi/4}}{\pi^{3/2}(k_1+\kappa\beta^2)^{1/2}x_1^{1/2}}} \qquad x_1 > 0$$

$$k_o = \frac{\omega}{\beta^2 c_\infty} \qquad \kappa^2 = k_o^2 - \frac{k_3^2}{\beta^2}$$

$$k_1 + k_o M = \frac{k_1}{\beta^2}$$

  - Notice the pressure jump is singular at the LE ($x_1 = 0$) and tends to zero at large distances ($x_1 \to \infty$)
    - Due to thin airfoil theory…
    - In reality, the velocity at the LE will be finite due to viscous effects and LE radius

$$k_1 = \frac{\omega}{U_\infty}$$

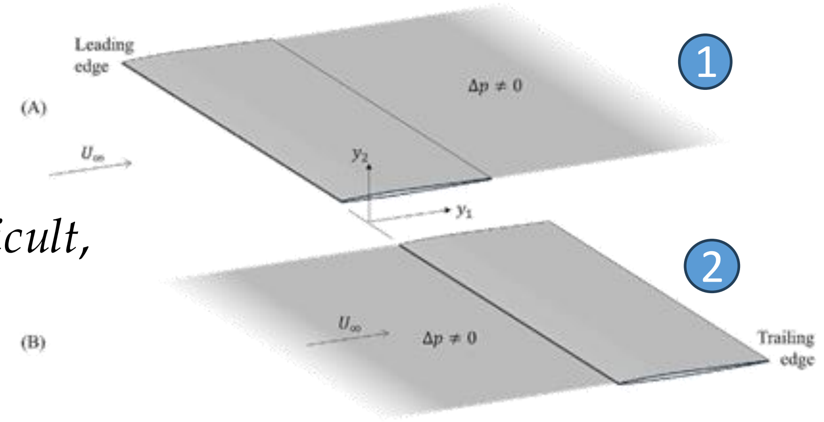


Amiet's leading- and trailing-edge solutions

LE solution

- For a finite chord blade, results need to be corrected for the effect of the TE and the Kutta condition
- TE scattering theory can be used to calculate a correction to the above that satisfies the BC at the TE
  - Yet, the TE result was for a semi-infinite flat plate, and so there will be a residual error that occurs at the LE
- While successive corrections can be calculated, let us first consider the **first-order TE correction** that is obtained by solving the TE problem with the incident pressure jump obtained from the LE solution
- Since the origin of the LE problem is at a distance c upstream of the TE, we re-write the solution of the LE problem with the origin at the TE:

$$\Delta\hat{p}^{(1)}(x_1, x_3) = \pi\rho_\infty \hat{a}_2 U_\infty \left\{ \frac{-2e^{i(\kappa - Mk_o)(x_1+c)+i\pi/4}}{\pi(k_1+\kappa\beta^2)^{1/2}\sqrt{\pi(x_1+c)}} \right\} e^{ik_3x_3}$$



- Carrying out the full Wiener–Hopf procedure on the function is *difficult*, but a **simplified result** is obtained in the **high frequency limit**, if the amplitude variation with x_1 is ignored, resulting in this approximation of the pressure near the TE

$$\Delta\hat{p}(x_1, x_3) \approx \pi\rho_\infty \hat{a}_2 U_\infty \left\{ \frac{-2e^{i(\kappa - Mk_o)c+i\pi/4}}{\pi(k_1+\kappa\beta^2)^{1/2}\sqrt{\pi c}} \right\} e^{i(\kappa - Mk_o)x_1 + ik_3x_3}$$

Amiet’s leading- and trailing-edge solutions
LE solution

❑ We can now perform the TE analysis (as previously performed using the Wiener–Hopf procedure), to find the TE correction as the addition pressure jump:

$$\boxed{\Delta\hat{p}^{(2)}\,(x_1, x_3) = \frac{-2\pi\rho_\infty \hat{a}_2 U_\infty e^{i(\kappa - Mk_0)(x_1+c)+i\pi/4}}{\pi(k_1+\kappa\beta^2)^{1/2}\sqrt{\pi c}}\left[(1-i)E_2(2\kappa|x_1|)-1\right]e^{ik_3x_3}}$$

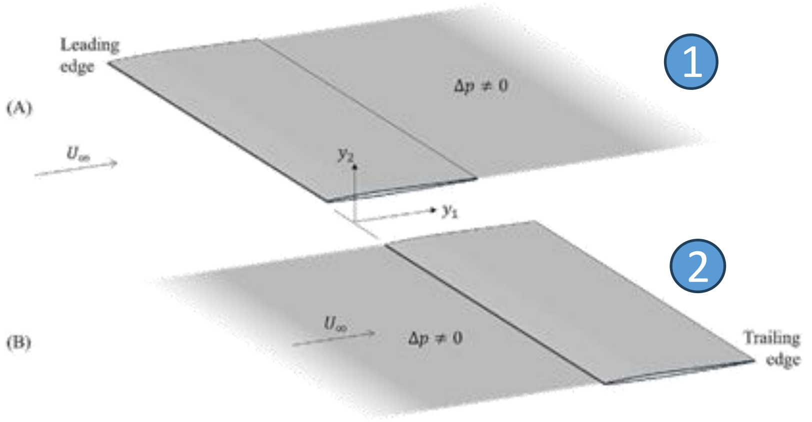
Modified Fresnel integral

$$E_2(x) = \int_0^x \frac{e^{i\zeta}}{\sqrt{2\pi\zeta}}\,d\zeta$$

❑ If the origin of the coordinate system is relocated back to the LE, we obtain the corrected non-dimensional surface pressure as:

$$\boxed{\Delta\hat{p}^{(1+2)}\,(x_1, x_3) = \pi\rho_\infty \hat{a}_2 U_\infty g^{(1+2)}(x_1, k_3, \omega, M)e^{ik_3x_3}}$$

$$\boxed{g_{\mathrm{TE}}(x_1, k_3, \omega, M) = \frac{-2e^{i(\kappa - Mk_0)x_1+i\pi/4}}{\pi(k_1+\kappa\beta^2)^{1/2}\sqrt{\pi c}}\left[\sqrt{\frac{c}{x_1}}+(1-i)E_2(2\kappa|x_1-c|)-1\right]}$$



➢ Although the above is just a first-order solution, it includes the major effects caused by the TE, and ensures that the Kutta condition is satisfied